

Lab 3: Multicycle Processor Digital Design and Computer Architecture: RISC-V Edition (Harris & Harris, Elsevier © 2021)

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TIME SPENT: 8 HOURS

GIT LINK:

[https://github.com/recepfatihyuksel/ELE432 Preliminary Work 3](https://github.com/recepfatihyuksel/ELE432_Preliminary_Work_3)

1. Processor Hierarchy Diagram

Figure 1 shows the required hierarchy. The top module contains the riscv processor and the unified memory. The riscv module contains the controller and datapath. The output signals WriteData, DataAdr, and MemWrite are tapped out for the testbench.

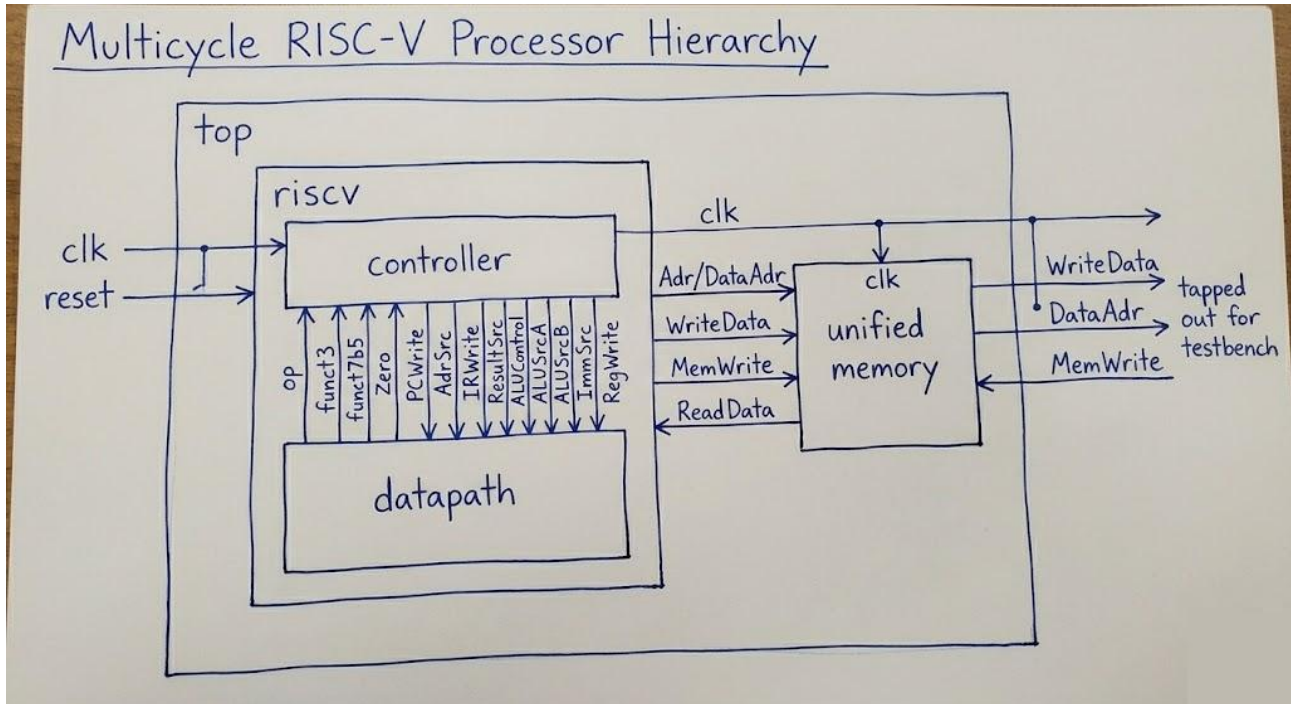


Figure 1. Multicycle RISC-V processor hierarchy.

2. SystemVerilog Implementation

Source code submission. The complete SystemVerilog files are submitted through the Git repository. The main processor file contains the top module, riscv module, controller, datapath, memory, and helper modules.

File	Purpose
processor.sv	Top-level processor, riscv, controller, datapath, memory, and helper modules.
riscv_testbench.sv	Simulation testbench and final success/failure condition.
memfile.txt	Machine code program loaded into memory.
run_rtl.do	Questa/ModelSim compile, simulation, and waveform script.
README_TR.txt	Short description of submitted files.

Top-level module interface used by the testbench:

```
module top(input  logic      clk, reset,
           output logic [31:0] WriteData, DataAdr,
           output logic      MemWrite);
```

Main modules included in processor.sv: top, riscv, controller, maindec, aludec, datapath, regfile, extend, alu, mem, flopr, flopenr, mux2, and mux3.

3. Expected Operation for the First Three Instructions

Program start. After reset, the first three instructions in memfile.txt are expected to execute as follows:

PC	Machine Code	Instruction	Expected Register Result
0x00	00500113	addi x2, x0, 5	x2 = 0x00000005
0x04	00c00193	addi x3, x0, 12	x3 = 0x0000000C
0x08	ff718393	addi x7, x3, -9	x7 = 0x00000003

Key signal table. The table below follows the multicycle state sequence for these instructions.

Step	PC	Instr	State	Result / ALUResult	Notes
3	0x00	00500113	S0: Fetch	0x00000004	Fetch first instruction and compute PC + 4.
4	0x04	00500113	S1: Decode	X	Decode addi x2, x0, 5.
5	0x04	00500113	S8: Executel	0x00000005	ALUResult = x0 + 5.
6	0x04	00500113	S7: ALUWB	0x00000005	Write result to x2.
7	0x04	00c00193	S0: Fetch	0x00000008	Fetch second instruction and compute PC + 4.
8	0x08	00c00193	S1: Decode	X	Decode addi x3, x0, 12.
9	0x08	00c00193	S8: Executel	0x0000000C	ALUResult = x0 + 12.
10	0x08	00c00193	S7: ALUWB	0x0000000C	Write result to x3.
11	0x08	ff718393	S0: Fetch	0x0000000C	Fetch third instruction and compute PC + 4.
12	0x0C	ff718393	S1: Decode	X	Decode addi x7, x3, -9.
13	0x0C	ff718393	S8: Executel	0x00000003	ALUResult = x3 + (-9) = 3.
14	0x0C	ff718393	S7: ALUWB	0x00000003	Write result to x7.

Table 1. Expected operation for the first three instructions.

4. Simulation Waveform and Testbench Result

Required waveform signals. The simulation waveform includes clk, reset, PC, Instr, state, SrcA, SrcB, ALUResult, DataAdr/Adr, WriteData, and MemWrite. The 32-bit signals are displayed in hexadecimal.

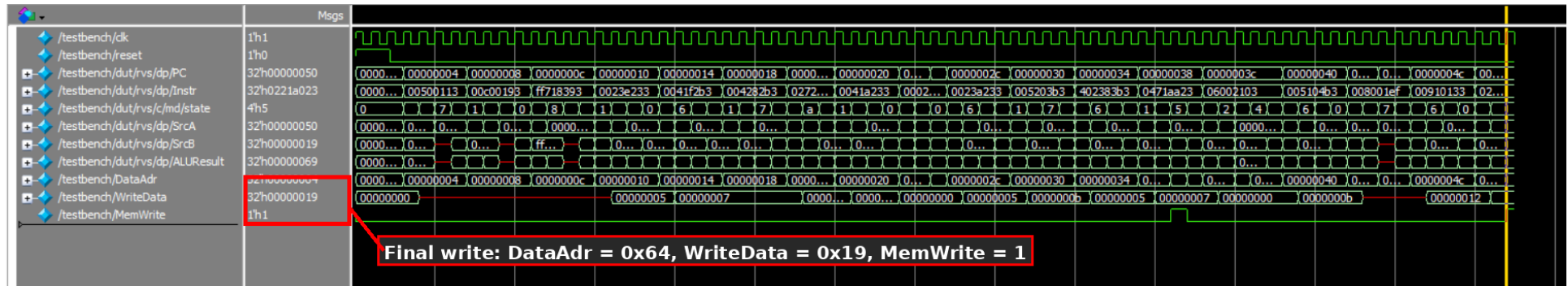


Figure 2. Required simulation waveform with final write condition highlighted.

Final write check. At the highlighted cursor position, DataAdr = 0x00000064 and WriteData = 0x00000019 while MemWrite = 1. Therefore, the processor writes decimal 25 to decimal address 100, which matches the testbench success condition.

Testbench message: Simulation succeeded: wrote 25 to address 100.